

Block Coding.

s, m
 $\checkmark$ $\boxed{10101101}$

n, R
 $\boxed{11101101}$ $\checkmark$

Redundant bit.

(i) Synchronization.
 (ii) Error Detection

(start bit or stop bit).

$\underline{mB/nB}$

$m < n$

$m > n$

3 steps
 (i) Division.
 (ii) Substitution
 (iii) Combination.

$m \text{ bit}$
 $\downarrow$
 $n \text{ bit}$

Where, $n > m$

$n > m$

$\underline{mB/nB}$

Division $\rightarrow$

$m \text{ bit}$	$m \text{ bit}$	$m \text{ bit}$	$m \text{ bit}$
1011	1101	1011	1010

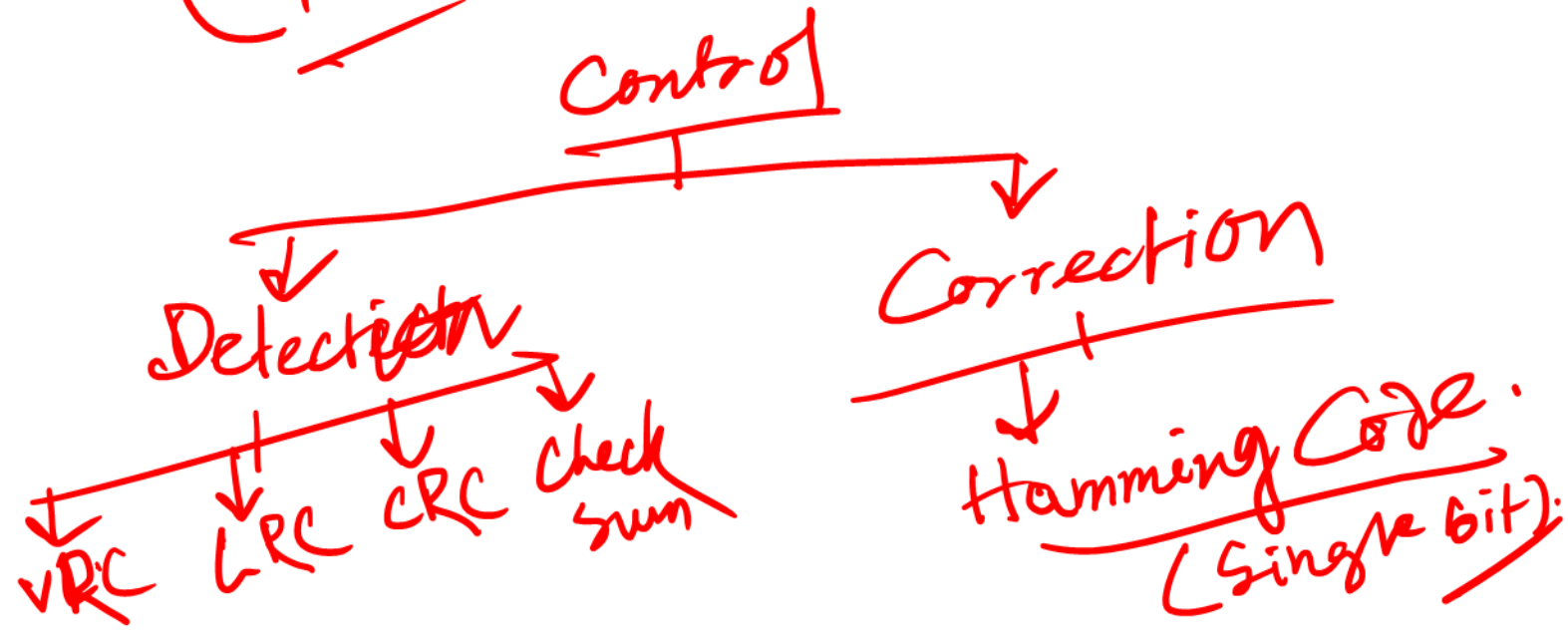
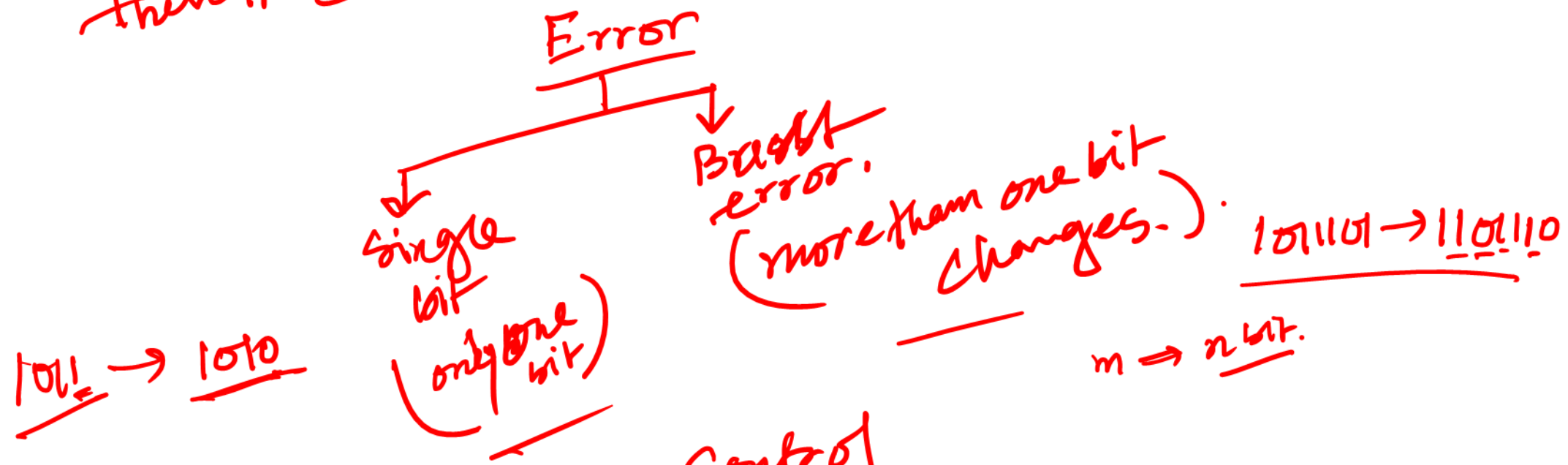
 Substitution $\Rightarrow mB/nB$, $n > m$
 Combination $\Rightarrow$

$m \text{ bit}$	$m \text{ bit}$	$m \text{ bit}$	$m \text{ bit}$
10110	11010	10110	10100

 $n \text{ bit}$ $n \text{ bit}$ $n \text{ bit}$ $n \text{ bit}$

Error Control:

Due to transmission impairments if any bit changes then it causes error.



VRC (Vertical Redundancy Check).

Sender side

1001 → Data



10010

send

through Network

10010 →

11000 → 0

10010
↓
11000

$\frac{1101}{1010} \rightarrow 1$
 $\frac{1010}{1010} \rightarrow 0$

101101 → 1

Accept
No error

Error.
NAK

Error

0

Even Parity?

10110

Receiver side.

Pros: → (i) Easy
(ii) Good for

single bit
error detection.

Cons: → If even no of bit
changes then error
can't be detected.

LRC → (Longitudinal Redundancy Check).

I → 73	→	1 0 0 1 0 0 1	→	1
N → 78	→	1 0 0 1 1 1 0	→	2
D → 68	→	1 0 0 0 1 0 0	→	3
I → 73	→	1 0 0 1 0 0 1	→	4
A → 65	→	1 0 0 0 0 0 1	→	5
		⊕		
		1 0 0 1 0 1 1	→	X

A side (Sender)

X	5	4	3	2	1
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B-side (Receiver side).

1 0 0 1 0 0 1	→	1
1 0 0 1 1 1 0	→	2
1 0 0 0 1 0 0	→	3
1 0 1 1 0 1 1	→	4
1 0 1 0 0 0 1	→	5
⊕		
1 0 0 1 0 1 1	→	X
0 0 0 0 0 0 0	→	0

Pros: (i) burst and single bit.
(ii) Better than VRC.

Cons: (i) Same position two or even no. of bit changes.
then it can't be detected.

if, zero (0) value.
then it is error free.
if, Non-zero value
then it is with error.

